

ASHEVILLE, NC, USA

WMO: 723150

Lat: 35.432N

Lon: 82.538W

Elev: 645

StdP: 93.81

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.5	(c) -7.0	(d) -17.8	(e) 0.8	(f) -4.4	(g) -15.5	(h) 1.1	(i) -3.2	(j) 12.2	(k) -1.7	(l) 11.1	(m) -1.8	(n) 5.5	(o) 340	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.1	21.5	29.9	21.2	28.8	20.9	23.2	28.2	22.6	27.3	22.1	26.5	3.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	17.8	24.9	21.2	17.2	24.5	20.7	16.6	24.0	72.0	28.2	69.7	27.4	67.5	26.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.6	7.7	-13.3	32.8	2.9	1.3	-15.4	33.7	-17.1	34.5	-18.7	35.2	-20.7	36.1		
			-14.2	24.3	2.7	1.2	-16.1	25.2	-17.7	25.9	-19.2	26.6	-21.2	27.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	3.1	5.1	8.7	13.6	17.8	21.7	23.2	22.8	19.7	14.0	8.5	4.9
	DBStd	8.14	5.49	5.00	4.90	4.10	3.43	2.36	1.92	1.90	2.96	4.03	4.41	4.84
	HDD10.0	731	223	148	84	14	1	0	0	0	0	11	80	170
	HDD18.3	2222	473	371	299	148	51	4	0	1	21	142	297	416
	CDD10.0	2057	8	11	43	122	243	350	410	398	290	136	33	13
	CDD18.3	507	0	0	0	6	36	104	152	140	61	8	1	0
	CDH23.3	3519	0	1	9	93	301	729	1028	896	402	58	2	0
	CDH26.7	894	0	0	0	8	54	189	303	246	88	7	0	0
Wind	WSAvg	2.8	3.7	3.7	3.6	3.3	2.6	2.2	2.0	1.8	1.9	2.4	3.0	3.3
Precipitation	PrecAvg	1498	125	116	137	120	109	141	139	143	131	96	107	135
	PrecMax	2150	297	237	256	221	253	269	361	256	491	223	219	259
	PrecMin	1008	33	49	51	54	37	40	49	26	24	7	38	59
	PrecStd	257	63	46	49	41	54	56	69	57	90	57	46	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.6	24.6	27.8	30.2	31.8	32.4	32.5	30.7	27.4	23.3	20.0
		MCWB	13.3	14.7	14.5	16.9	19.4	21.1	22.3	22.0	19.8	19.8	14.9	14.2
	2%	DB	16.4	18.1	22.0	26.0	28.2	30.2	30.9	30.7	29.0	24.9	20.5	17.6
		MCWB	12.6	12.0	13.4	16.0	18.9	20.9	22.0	21.7	19.9	17.4	13.3	13.5
	5%	DB	14.3	16.0	19.7	24.1	26.6	28.9	29.7	29.3	27.5	23.0	18.4	15.1
		MCWB	10.8	11.0	12.3	15.0	18.5	20.5	21.9	21.5	19.7	16.5	13.2	11.6
	10%	DB	11.8	13.6	17.3	22.0	24.9	27.5	28.5	27.9	25.9	21.2	16.5	13.0
		MCWB	9.1	9.1	11.2	14.1	17.7	20.0	21.5	21.0	19.2	15.7	12.4	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	16.2	16.3	19.1	21.4	23.0	24.2	23.9	22.4	21.5	18.0	16.6
		MCDB	17.4	19.9	20.9	25.3	26.7	28.5	29.5	28.8	26.9	25.3	19.9	18.8
	2%	WB	13.6	14.3	15.1	17.6	20.4	22.3	23.4	23.1	21.6	19.7	16.1	14.5
		MCDB	15.5	16.9	19.8	23.0	25.4	27.5	28.4	28.0	25.9	22.4	18.3	16.4
	5%	WB	11.5	12.1	13.9	16.6	19.7	21.7	22.8	22.5	20.8	18.2	14.8	12.5
		MCDB	13.6	14.5	17.9	21.2	24.3	26.6	27.6	27.0	24.7	21.0	17.3	14.3
	10%	WB	9.3	9.9	12.5	15.5	18.9	21.0	22.2	21.9	20.2	17.0	13.1	10.5
		MCDB	11.2	12.5	15.9	19.8	23.2	25.7	26.7	26.1	23.7	19.7	15.5	12.4
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.0	11.8	12.7	11.9	10.9	10.2	10.2	10.6	11.9	12.1	10.3
		MCDBR	14.1	15.2	15.7	16.2	13.9	13.1	11.9	12.2	13.0	14.2	15.0	13.4
		MCWBR	9.4	9.5	8.5	7.6	6.1	5.1	4.7	4.7	5.2	7.1	8.8	8.9
	5% WB	MCDBR	11.2	11.9	12.8	12.7	11.6	11.1	10.6	10.4	10.3	10.0	10.8	10.7
		MCWBR	8.6	9.0	8.1	6.8	5.5	4.8	4.5	4.4	4.6	5.9	7.9	8.3
Clear-Sky Solar Irradiance	taub	0.294	0.303	0.333	0.373	0.426	0.452	0.493	0.490	0.386	0.352	0.313	0.292	
	taud	2.544	2.512	2.455	2.344	2.211	2.172	2.087	2.074	2.350	2.440	2.515	2.579	
	Ebn at Noon	917	946	939	911	862	836	799	793	870	874	883	900	
	Edh at Noon	79	91	105	124	144	149	161	160	116	97	81	72	
All-Sky Solar Radiation	RadAvg	2.59	3.22	4.26	5.28	5.80	6.10	5.67	5.31	4.60	3.84	2.89	2.23	
	RadStd	0.16	0.34	0.27	0.40	0.43	0.42	0.39	0.25	0.41	0.40	0.25	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	-116	+167	+88
Regional (41 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+100

Nomenclature: See separate page